AP CALCULUS AB

Unit 1 • Lesson 1-9 • Notes (filled in)

Lesson 1-9 • IVT, EVT, and MVT

Notes (filled in)

Learning goals

- Choose the right existence theorem for a question.
- Check the hypotheses of the IVT, EVT and MVT from formulas, tables and graphs.
- Write complete AP-style justifications.

Vocabulary. existence theorem, hypothesis, conclusion, justification.

1. Which theorem?

Key idea

	Hypothesis	Conclusion
IVT	continuous on $[a,b]$	$f(c) = k$ for every k between $f(a)$ and $f(b)$
EVT	continuous on $[a,b]$	f has an absolute maximum and minimum on $[a,b]$
MVT	continuous on $[a,b]$ and differentiable on (a,b)	$f'(c) = \frac{f(b)-f(a)}{b-a}$ for some c in (a,b)

Questions about a value of f point to the IVT; about the largest or smallest value, the EVT; about a rate (value of f'), the MVT.

Example 1 • A table (AP style)

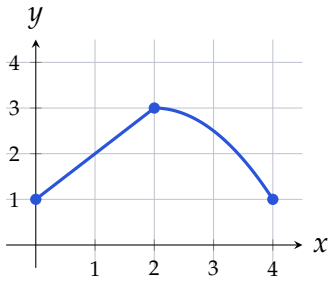
A differentiable function f has the values below.

x	0	2	5	7	10
$f(x)$	4	-2	3	9	5

(a) Must $f(c) = 0$ for some c in $[0,10]$? (b) Must $f'(c) = 3$ for some c in $(5,7)$? (c) Must f have an absolute maximum on $[0,10]$? (d) Must $f'(c) = 0$ for some c in $(0,10)$?

f is differentiable, hence continuous everywhere.
(a) Yes: on $[0,2]$, $f(0) = 4 > 0 > -2 = f(2)$; by the IVT there is c with $f(c) = 0$ (also on $[2,5]$).
(b) Yes: $\frac{f(7)-f(5)}{7-5} = \frac{6}{2} = 3$; by the MVT, $f'(c) = 3$ for some c in $(5,7)$.
(c) Yes: f is continuous on the closed interval $[0,10]$; by the EVT it has an absolute maximum.
(d) Yes. $f(0) = 4$, and since $3 < 4 < 9$, the IVT gives a b in $(5,7)$ with $f(b) = 4$. On $[0,b]$ the MVT gives $f'(c) = \frac{f(b)-f(0)}{b-0} = \frac{4-4}{b} = 0$ for some c in $(0,b)$.

Example 2 • Hypotheses fail



The graph of f on $[0,4]$ is shown (a corner at $x = 2$). Which theorems apply on $[0,4]$?

f is continuous on $[0,4]$: the IVT and EVT apply. f is not differentiable at $x = 2$ (corner), so the MVT does not apply on $[0,4]$ (it does apply on $[2,4]$).

AP exam tip

Never apply a theorem without stating why its hypotheses hold. On tables, "f is differentiable, so it is continuous" covers both continuity and differentiability.

Check your understanding

1. "Is there a time when the temperature was exactly 70°?" uses the __IVT__.
2. "Is there a time when the temperature was rising at 2° per hour?" uses the __MVT__.